

Classification of AIS Ship Trajectories using Feature Extraction and Class Imbalance Handling

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1 Problem Statement and Background Information

The goal of this project is to build an ML model that can classify the type of vessel activity into two categories: fishing and non-fishing, using Automatic Identification System (AIS) data. AIS data provide real-time information on vessel motion, including speed, position, course, and other vessel attributes. Such classification can be extremely useful in monitoring maritime activities, managing fisheries, and detecting any illegal fishing activities.

One of the main problems with the current project (Pedroche, 2020) is how to ensure the generalization of behaviors learned by the algorithm without simply memorizing the information about the particular vessel. It was discovered that during the analysis stage, some specific features, like the identifier of the vessel, can cause data leakage, thus allowing the model to simply memorize individual vessels rather than learn about their behavior.

To solve this problem, features that easily identify the vessel, such as low speeding, loitering, high turn rate behavior, and others that lead to data leakage, were removed, and a range of machine learning algorithms were considered and compared, which include Random Forest, Logistic Regression, and XGBoost. The exploratory data analysis, t-test, and DBSCAN and visualizations are performed in order to analyze the data better.

The ultimate objective is to develop an accurate classification model for a vessel's activity in terms of fishing using trajectory information.

2 Dataset Overview, Cleaning, and Preparation

2.1 Variables Description

Table 1 describes the variables in the data.

Table 1: Description of AIS Dataset Variables

Variable	Type	Description
cog	float	Course Over Ground (degrees)
sog	float	Speed Over Ground (knots)
beam	float	Width of the vessel (meters)
callsign	object	Vessel call sign identifier
cargo	object	Type of cargo carried by the vessel
heading	float	Direction vessel is pointing (degrees)
imonumber	float	Unique IMO ship identification number
latitude	float	Vessel latitude coordinate
length	float	Length of the vessel (meters)
longitude	float	Vessel longitude coordinate
mmsi	int	Maritime Mobile Service Identity (unique vessel ID)
name	object	Name of the vessel
navstatus	object	Navigation status (e.g., underway, anchored)
timeoffix	int	Timestamp of AIS message
vesseltype	object	Type/category of vessel
date	object	Date of recorded observation

2.2 Database Process

In the initial phase, the dataset downloaded from Kaggle in CSV format is stored in a MySQL database. For this purpose, we can make use of SQL commands and databases that facilitate us in storing and loading data into tables in a MySQL database.

We establish the connection to the MySQL database using `mysql.connector` in Jupyter Notebook. After connecting to the database, SQL queries can be used to extract the required data from it.

After retrieving the data, it is loaded into a pandas DataFrame using the `read_sql` function. This facilitates further processing of data, such as cleaning data, dealing with missing values, selecting appropriate features, and performing transformations.

2.3 Data Collection and Preprocessing

The dataset used in this project is a real AIS vessel tracking dataset that includes information such as speed over ground, course over ground, latitude, longitude, dimensions of the vessels, type of cargos carried by the vessel, navigation status, among other vessel-identifying parameters. Such data helps us understand the vessel’s motion behavior as well as its location. Fishing vessels have different behaviors in comparison with non-fishing vessels, like speed, movement behavior, and the specific geographical regions in which they operate. While fishing vessels travel at a low speed and follow random patterns, cargo vessels, which are non-fishing vessels, travel at a constant speed along specific routes.

This dataset is a real dataset taken from [kaggle.com](https://www.kaggle.com) and comprises 17 features and 103,997 observations. The data was saved in the MySQL database and imported into the Jupyter notebook using SQL queries.

While preprocessing the dataset, the initial step involved loading the data in the pandas DataFrame and examining it to identify and remove unwanted data elements. In particular, there existed an extra column named `MyUnknownColumn` in the dataset, which had to be removed as it did not offer any important insight about the data. All

the columns' names were converted into lower case to ensure consistency of the naming convention in the dataset. Before building the models, features that give identification information were removed to avoid data leakage.

The dataset was evaluated for missing values and inconsistencies. From the preliminary evaluation, the key features used for modeling did not contain significant missing values, so no major imputation was required. However, after feature engineering, a simple imputation technique was added to the pipeline during the model-building process. Data types were also reviewed to ensure that numerical features such as speed, latitude, and longitude were correctly formatted for analysis.

Variables like cargo type, type of vessel, and navigational status were converted to numeric representation through the technique of one-hot encoding, making them usable for machine learning algorithms. There were outliers present in the data set. These outliers were identified using DBSCAN and box plots and were addressed before building the models. The data was split into 80% for training and 20% for testing.

For better clarity of data visualization through plotting, the outliers in data were excluded, and the plots were generated on the basis of filtered data, keeping the original untouched for further use in the modeling process.

Class imbalance exists in the data; there are more non-fishing instances than fishing instances. To address the class imbalance between fishing and non-fishing states, the Synthetic Minority Oversampling Technique (SMOTE) was used. The technique works by creating synthetic data points that lie between two actual data points from the minority class, thus allowing for sensitivity to the fishing signal while avoiding overfitting.

All in all, the entire preprocessing process ensured that the data set would be ready for multiple machine learning models.

2.4 Feature Engineering

Feature engineering was performed to generate additional features. The kinematics factors such as acceleration, rate of turn, and change of direction were extracted using AIS data to determine the dynamics of vessel movement. These factors were then used to create other behavioral factors such as slow movement, high turning activity, and loitering patterns, which are usually indicative of fishing activities. Also, spatial features such as distance traveled and position-based speeds are some of the spatial factors created to improve on the description of movement. This multi-level feature engineering approach enables effective differentiation between fishing and non-fishing vessel behavior.

3 Exploratory Data Analysis

Exploratory data analysis was performed to understand patterns and relationships within the dataset using visualizations and summary statistics.

3.1 Spatial Spread of the Class

Figure 1 shows the spatial distribution of the ships. Fishing class exhibits geographical restriction compared to non-fishing class, which has global distribution. This implies that there is a geographic influence on the results of classification where there is regional favoritism to the Western Hemisphere in favor of the fishing class.

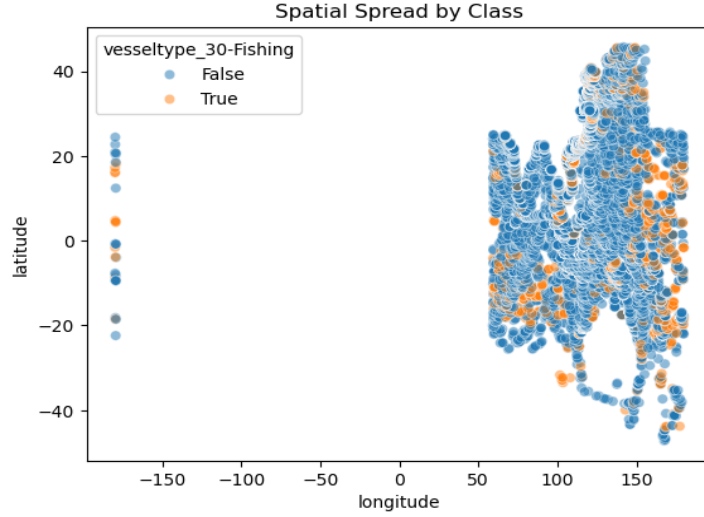


Figure 1: Spatial Spread by Class

3.2 Summary Statistics of the Direction

Table 2 shows the summary statistics of the coordinates of the trajectory. From the table, the mean latitudes and longitudes (approximately 15.28° and 114.90° , respectively) imply a high concentration of vessels in the Eastern Hemisphere, including areas like Southeast Asia and other water bodies near it. From the comparison of median latitude (21.94°) and the mean latitude, it can be concluded that there is a skew towards northern latitudes.

From the standard deviation values, it can be noted that there is a high geographical dispersion in the distribution of the vessels because of the larger number. It implies that they do not occur in close proximity to each other. The minimum and maximum values illustrate the wide range covered by the dataset, as the vessels span a latitudinal range from approximately 47° South to 45° North and a longitudinal range almost entirely (-180° to 180°). Nevertheless, the interquartile range (25% - 75%) indicates that the concentration of vessel occurrences is much denser at latitudes ranging from 1.6° to 26.0° and longitudes between 108.4° and 120.7° .

Thus, the presented data includes both extensive coverage of the globe and high concentrations in certain regions.

Table 2: Summary Statistics for Geographic Coordinates

Statistic	Latitude	Longitude
Count	80266	80266
Mean	15.285	114.899
Std Dev	15.440	18.448
Min	-47.222	-179.996
25%	1.615	108.425
Median (50%)	21.943	115.352
75%	25.987	120.743
Max	45.693	179.995

3.3 Distribution of Numeric Variables

Figure 2 shows the distribution of the numeric features. The numeric features provide insight into certain characteristics of the AIS dataset. As expected, most features show

non-Gaussian and skewed distributions, which is quite normal for data related to marine transportation. In particular, the SOG feature appears highly right-skewed, meaning that ships mainly travel at low speed levels and have very few episodes of high-speed operation. COG and heading, in turn, show roughly uniform distribution within the range of 0-360 degrees, showing there is no clear dominant direction of ships' movement; yet, heading features slightly irregular peaks, which can either be attributed to possible measurement errors or indicate that vessels are sometimes static. Length and beam distributions of vessels demonstrate strong right skewness, suggesting that mostly medium to small vessels are considered. Lastly, latitudes and longitudes display definite geographical clustering, confirming that ships are active only in some distinct zones. In addition, the hour attribute is highly concentrated on some particular values. Thus, it is possible to observe that the data shows such characteristics as skewness, outliers, and geographical and temporal clustering.

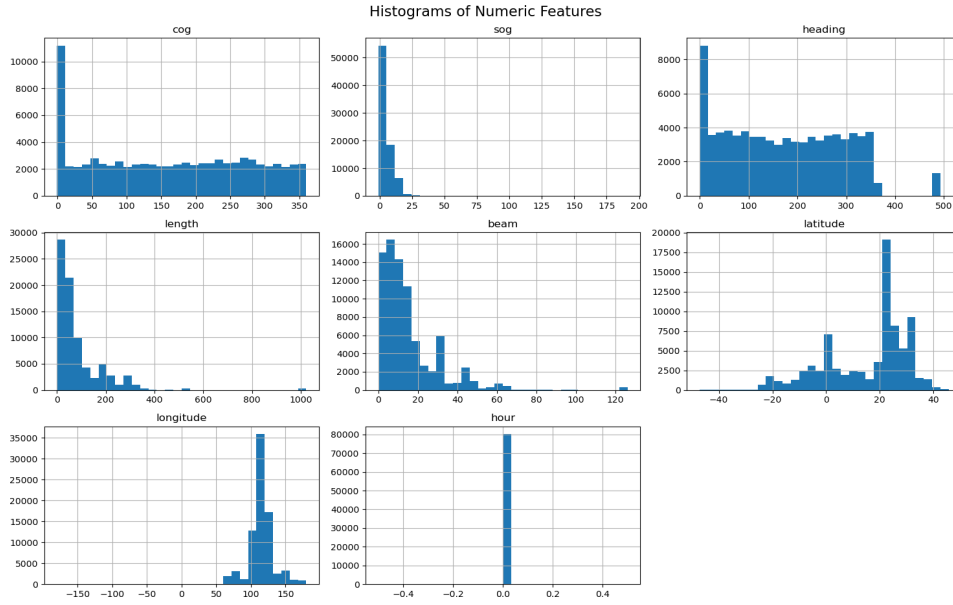


Figure 2: Histogram of Numeric Variables

Figure 3 shows the distribution of the numeric variables using boxplots. The plot shows that there is considerable variability and many outliers in the data. For example, the SOG, length, and beam variables have a high degree of right skewness and contain many outliers. This means that, although most of the vessels have normal measurements, there are still some that have unusually high figures. The size of the vessel, especially the length, has many deviations and extreme outliers, which may be attributed to large ships such as oil tankers and freighters. COG and heading are highly variable and show many deviations in their distribution because vessels take different paths during navigation. Geographical variables such as latitude and longitude have some variability and outliers. This implies that although most vessels operate in specific locations, some operate in different areas. The hour variable seems to have minimal variability, and there are few variations in its distribution.

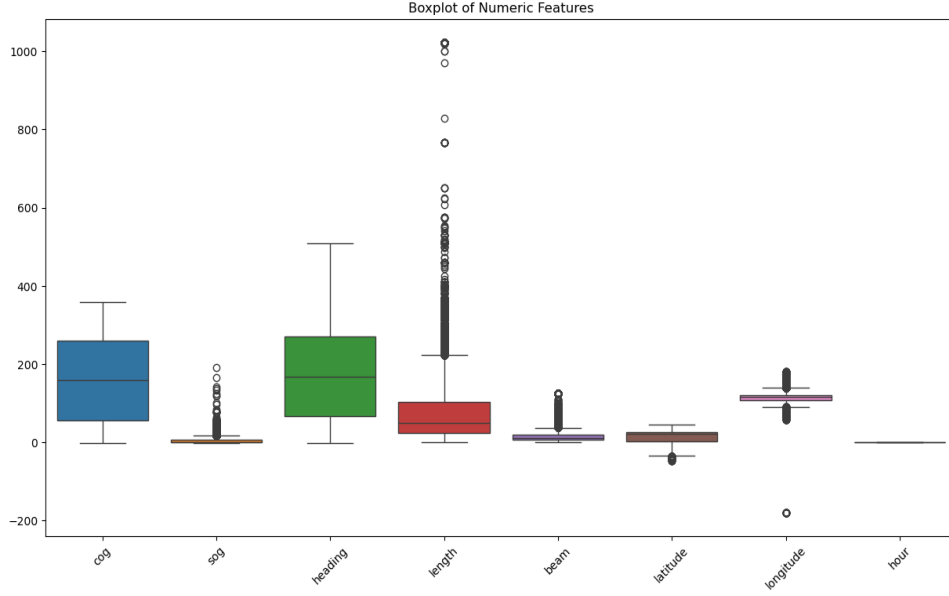


Figure 3: Box Plot of Numeric Variables

3.4 Behavioral Analysis (SOG vs COG)

Figure 4 shows the scatter plot of movement behavior of the ships. From the SOG-COG scatterplot, there are clear patterns in terms of vessel behavior regarding its movement. First, we see an accumulation of observations at low speeds, i.e., below 20 knots. In other words, vessels tend to move slowly rather than fast, since there is a high number of observations made on low speed levels. The course over ground varies from 0 to 360 degrees; thus, it means that ships tend to move in various directions while their speed is low. This behavior is typical for maneuvering, anchoring, and fishing activities. With increasing speeds, the number of observations becomes lower, and the density of COG also becomes smaller. This pattern shows us that the vessel moves rarely and faster. Moreover, the few points on the graph that have extremely high values are outliers that can be caused by data error or some special ship operation activities.

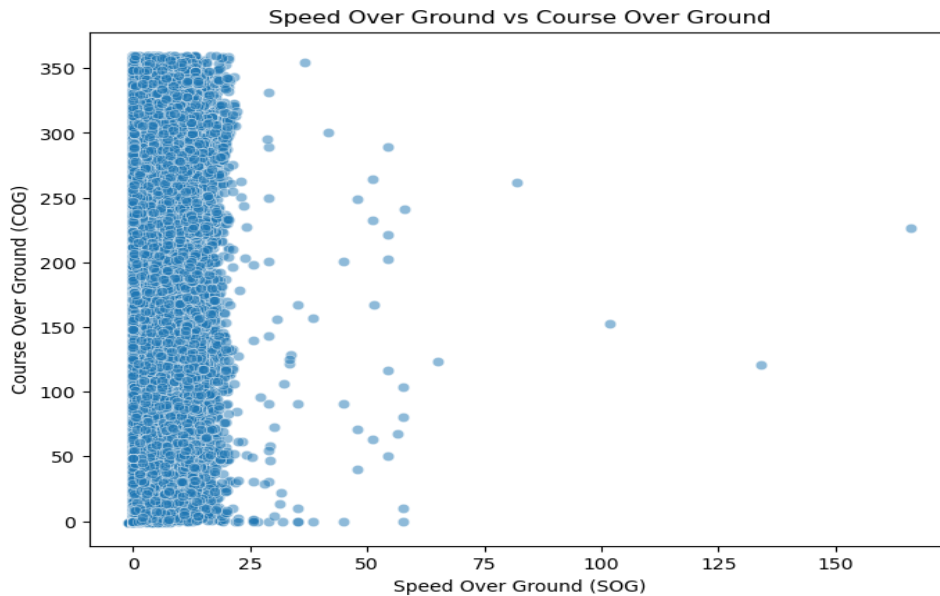


Figure 4: Behavior Analysis Using Course over Ground and Speed over Ground

3.5 Speed Over Ground Distribution by Fishing Activity

Figure 5 shows the distribution of the class of the target variable with respect to speed over ground. The plot indicates that active fishing has a unique speed characteristic in terms of its high median speeds (2.5 knots) relative to times that fishing activities were not being carried out. Although both types of activity have comparable high transit speeds of 30 knots, active fishing speeds are mostly observed within a range of 1-7 knots.

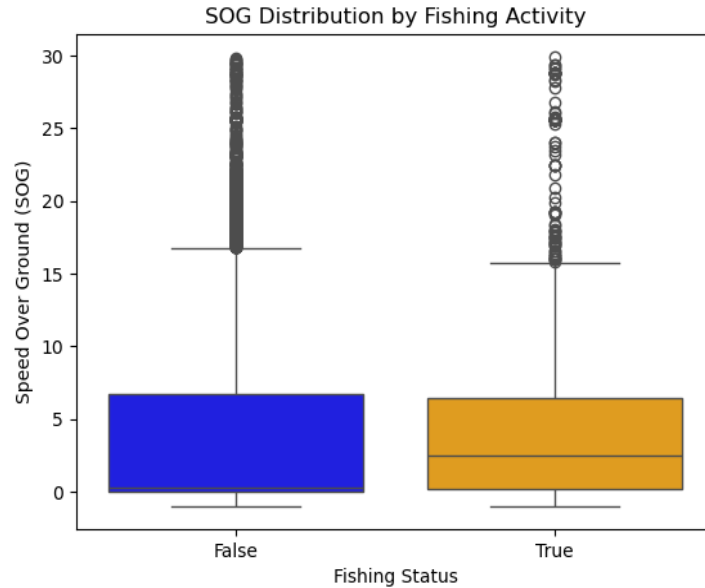


Figure 5: Box Plot of Target Feature Using SOG

3.6 Correlation Analysis of Features

The correlation plot of the features is displayed in figure 6. It can be noted from the correlation matrix that there is a high dependency between the orientation of the vessels (COG and heading) and expected relationships between speed and distance. Nevertheless, it can be observed that there is no high linear dependency between the target variable (is_fishing) and each of the features individually.

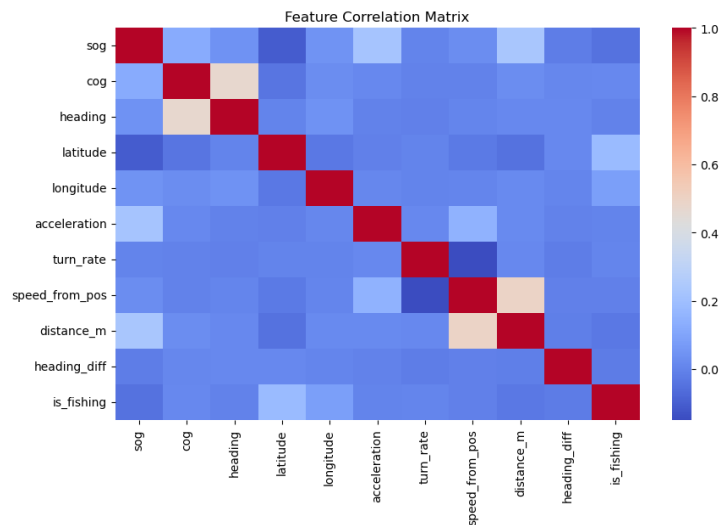


Figure 6: Correlation of Features

4 Classical Statistical Techniques

Table 3 provides the independent t-test results of the significance of input features in classifying ships. From the t-test results, latitude/longitude and SOG appear to be the main statistical discriminants when it comes to fishing status of the ships. Although heading_diff is shown to vary considerably among states, acceleration and turn_rate are not observed to have significant differences in their means, suggesting they may be less reliable as standalone indicators for fishing.

Table 3: T-test Results Comparing Fishing vs. Non-Fishing States

Feature	p-value	Significance
Latitude	$< 1.00e - 100$	***
Longitude	4.06e-69	***
SOG	3.36e-33	***
Heading Diff	1.26e-06	***
Distance (m)	1.93e-06	***
COG	8.95e-03	**
Speed from Pos	5.99e-02	n.s.
Heading	3.41e-01	n.s.
Turn Rate	3.69e-01	n.s.
Acceleration	9.51e-01	n.s.

Significance levels: *** $p < 0.001$, ** $p < 0.01$, n.s. = not significant

Table 4 is the analysis of normality, skewness, and kurtosis of the features. The distribution results show that the data is extremely skewed and leptokurtic, meaning that there are heavy tails on either side of the mean. This phenomenon is especially observed in the motion-related variables speed_from_pos and turn_rate. In such situations, classical assumptions may be violated, and therefore, appropriate methods should be used.

Table 4: Normality, Skewness, and Kurtosis Analysis

Feature	Skewness	Kurtosis	Shapiro-Wilk (p)	Normal?
Speed from Pos	81.85	8394.02	4.60×10^{-94}	False
Distance (m)	27.04	949.61	4.86×10^{-92}	False
Acceleration	18.75	1435.39	5.25×10^{-88}	False
SOG	3.04	39.55	4.67×10^{-72}	False
Turn Rate	-84.07	12 729.25	9.10×10^{-71}	False
Heading Diff	0.01	1.30	5.70×10^{-57}	False
Longitude	-1.02	20.01	1.30×10^{-51}	False
Latitude	-0.61	-0.66	1.33×10^{-44}	False
COG	0.06	-1.29	1.08×10^{-43}	False
Heading	0.26	-0.80	2.45×10^{-37}	False

Note: All features significantly deviate from a normal distribution ($p < 0.05$).

5 Behavioral Clustering Analysis Using Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

A DBSCAN algorithm was conducted for SOG and Turn Rate in order to group normal patterns from anomalies. This method effectively identified 98.7% of the data points under one main cluster (Cluster 0). There were 513 data points (1.2%) which were recognized as noise or outliers (Cluster -1), with very high values of turn rate and SOG shown in figure 7. The outliers may be errors coming from faulty sensors or sudden spikes in the GPS telemetry.

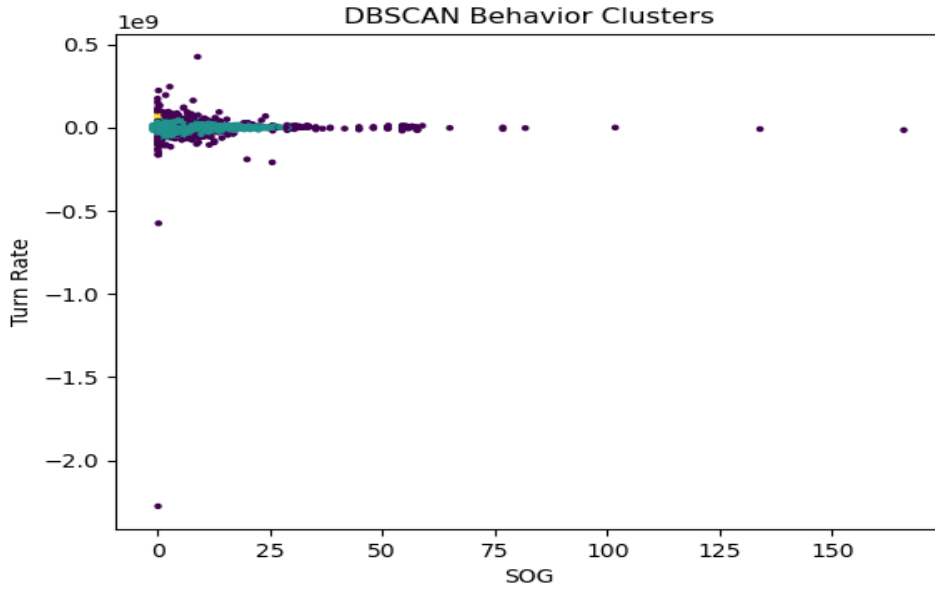


Figure 7: DBSCAN Using Turning Rate and SOG

6 Models and Comparisons

In this experiment, a number of supervised machine learning classifiers have been developed to predict whether the vessel is engaging in fishing activity or not using information obtained from AIS. First of all, preprocessing of data was performed by cleaning it from extreme sensor noise and transformation into features reflecting time-based, motion-based, and kinematic properties of vessels' trajectories. Additionally, categorical navigation and vessel-type characteristics were encoded, while missing data has been dealt with through imputation in the same preprocessing pipeline. Three classifiers have been trained, namely Logistic Regression, Random Forest Classifier, and XGBoost Classifier, with the latter two representing ensemble learning approaches of bagging and gradient boosting, respectively. To ensure robust evaluation, stratified k-fold cross-validation has been applied to all classifiers to ensure the same distribution of samples across classes in each iteration. Scaling and SMOTE algorithms were integrated into the pipeline of all three classifiers.

Table 5 provides the comparison of the three models using five metrics, namely, accuracy, precision, recall, F1-score and Receiver Operation Characteristic-Area Under the

Curve (ROC-AUC). From the results shown above, the Logistic Regression model shows less optimal predictive performance than ensemble models. The model’s low precision and F1 scores suggest that the Logistic Regression approach fails to predict nonlinear vessel trajectory movements effectively. However, the Logistic Regression model performs optimally when considering its recall and ROC-AUC values since it can detect most of the fishing instances.

On the other hand, the Random Forest and XGBoost algorithms demonstrate optimal predictive performance in all evaluation criteria due to their ability to handle nonlinear associations and temporal dependencies in AIS data. While the Random Forest model balances the precision-recall curve, the XGBoost algorithm enhances the model’s accuracy and precision.

Table 5: Performance Comparison of Machine Learning Models for AIS-Based Fishing Detection

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.753	0.234	0.867	0.369	0.862
Random Forest	0.955	0.696	0.810	0.748	0.974
XGBoost	0.957	0.728	0.780	0.753	0.972

Figure 8 shows the comparison of the three models from the cross-validation using accuracy, precision, recall, F1-score, and AUC. From the results obtained from the comparison, XGBoost emerges as the most superior algorithm with an F1-Score of 0.75 and ROC-AUC of 0.97. The statistically significant difference in performance between the two groups (ensemble techniques and Logistic Regression with an F1-Score of 0.37) confirms the choice of using non-linear algorithms based on decision trees due to their capability to deal with extreme outliers and skewed distributions. The high recall rate of 0.78 ensures that the majority of fishing events are successfully identified, while maintaining a precision level (0.73) suitable for practical monitoring applications.

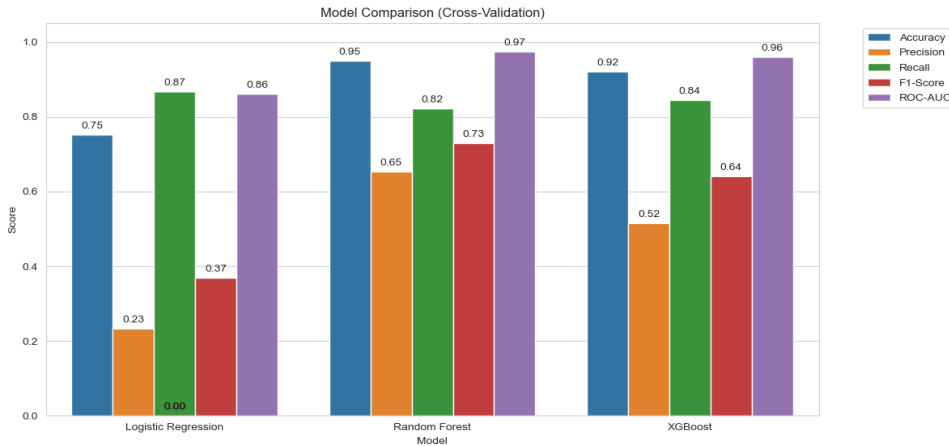


Figure 8: Model Comparison From Cross-Validation

Figure 9 shows the train and test performance of the three models. Upon conducting an analysis of the generalization gap, it can be deduced that XGBoost is the best-performing algorithm since it displays consistency across the training and testing stages, showing a consistent 5% variance. Although Random Forest displayed a greater training accuracy than other models, the high generalization gap of 18% highlights the issue of overfitting. On the other hand, Logistic Regression showed zero generalization gap, but it

failed to demonstrate predictive relevance (0.55), which shows high bias. Consequently, XGBoost is recommended for operational use due to its superior ability to generalize movement patterns without memorizing telemetry noise.

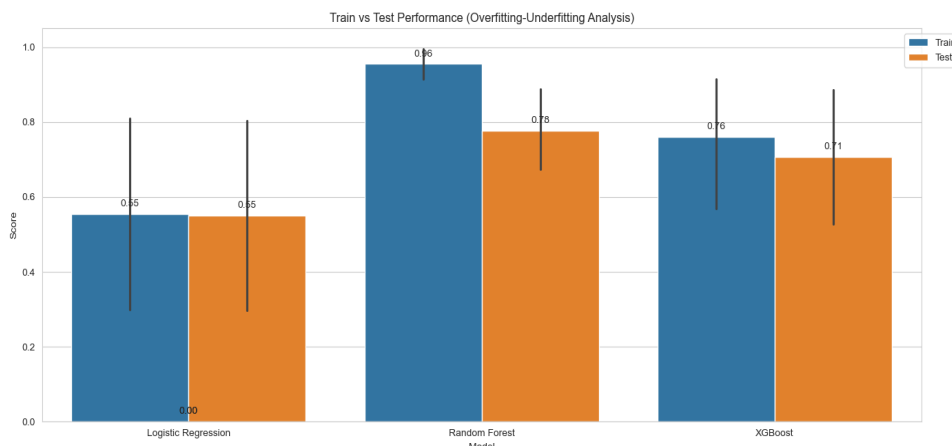


Figure 9: Underfitting and Overfitting Analysis

Figure 10 is the confusion matrix to show how the three models predicted the classes. The confusion matrix shows that the ensemble approaches (XGBoost and Random Forest) are far superior to the Logistic Regression model when it comes to reliability in practice. For example, the number of false positives was lowered by more than 90 percent by using XGBoost (31 as opposed to 335), thus reducing the chances of "false alarms." Both of the ensemble approaches showed good performance in detecting the minority fishing class, achieving about 71 percent classification of active fishers while preserving high accuracy rates.

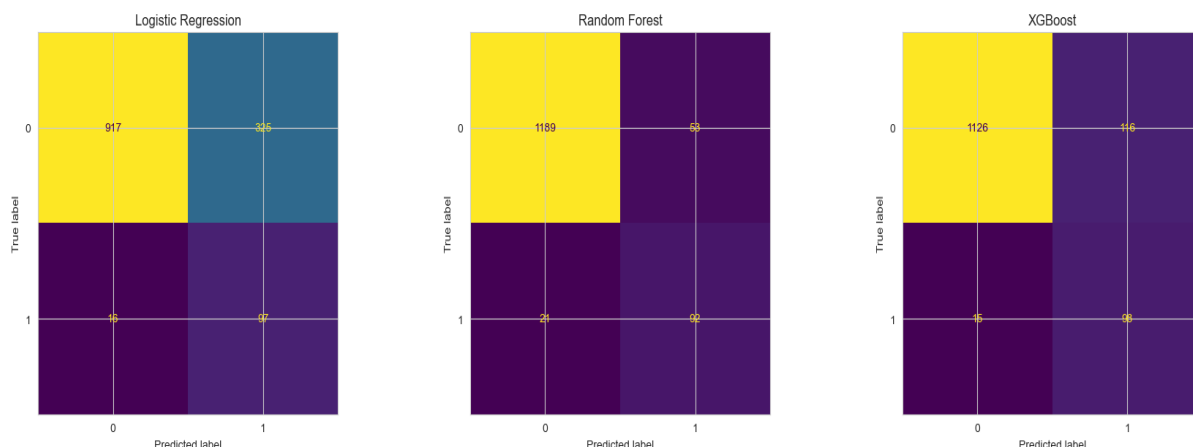


Figure 10: Confusion Matrix of the Three Models

Figure 11 provides the information on the feature importance for the random forest model. The ranking of feature importance reveals that vessel size (width and length) is the most significant variable for predicting fishing behavior, implying a clear association between vessel size and the behavior exhibited. After physical parameters, speed-related variables (Speed Over Ground, SOG, and estimated speed) become the second set of primary features, thus supporting the concept of a "speed signature." It is intriguing to note that location parameters (latitude/longitude) come in third, signifying that although fishing happens at particular locations, vessel physical characteristics and speed are more reliable for classification.

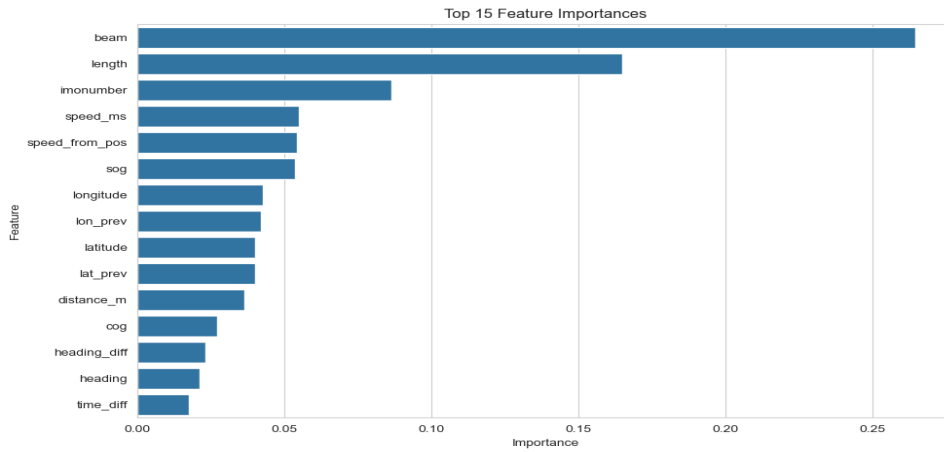


Figure 11: Feature Importance for Random Forest Model

7 Conclusion

In conclusion, a powerful machine learning pipeline was established to categorize AIS ships' tracks into fishing and non-fishing classes. With the use of advanced ensemble models such as Random Forest and XGBoost, the current study took into account the peculiarities of maritime data, which include non-linear tracks and outliers detected using the DBSCAN algorithm (p. 9). Moreover, the utilization of SMOTE made it possible to balance class distribution and avoid overfitting the minority fishing class (p. 3). From the analysis, Logistic Regression exhibited bias, Random Forest exhibited overfitting ,and XGBoost is well fitted (optimal)(p. 10-11).

Overall, XGBoost proved to be the best-performing algorithm from the cross-validation, as it produced an F1-Score of 0.75 and an ROC-AUC of 0.97 (p. 10) and testing accuracy of 0.71 (p. 11) Additionally, analyzing feature importance indicated that parameters like vessel size and speed are better determinants of fishing activity compared to geographical location (p. 11-12).

References

- [1] Sánchez Pedroche, D., Amigo, D., García, J., & Molina, J. M. (2020). Architecture for Trajectory-Based Fishing Ship Classification with AIS Data. *Sensors*, 20(13), 3782. <https://doi.org/10.3390/s20133782>
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